

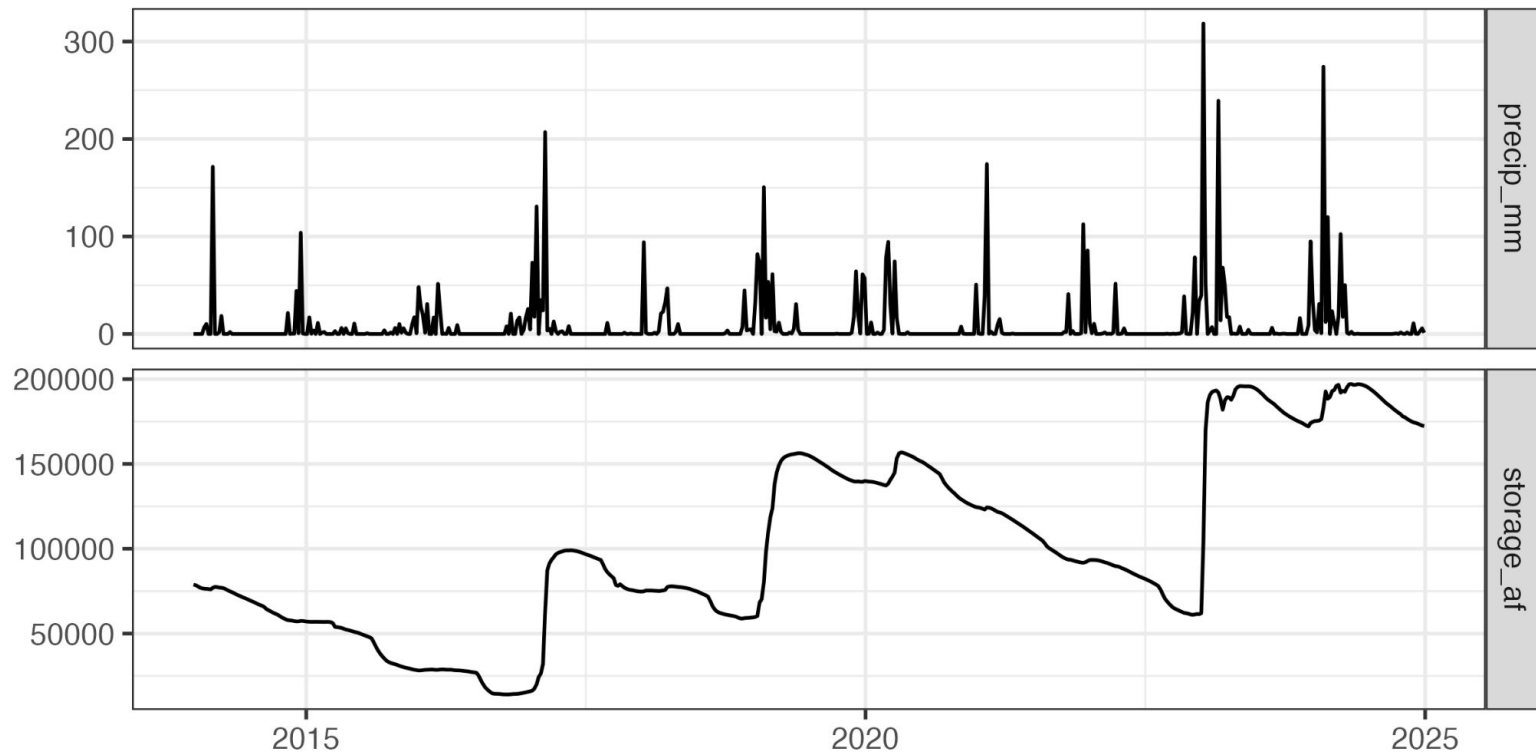
Week 8 Lecture 2:

Spatial and temporal autocorrelation

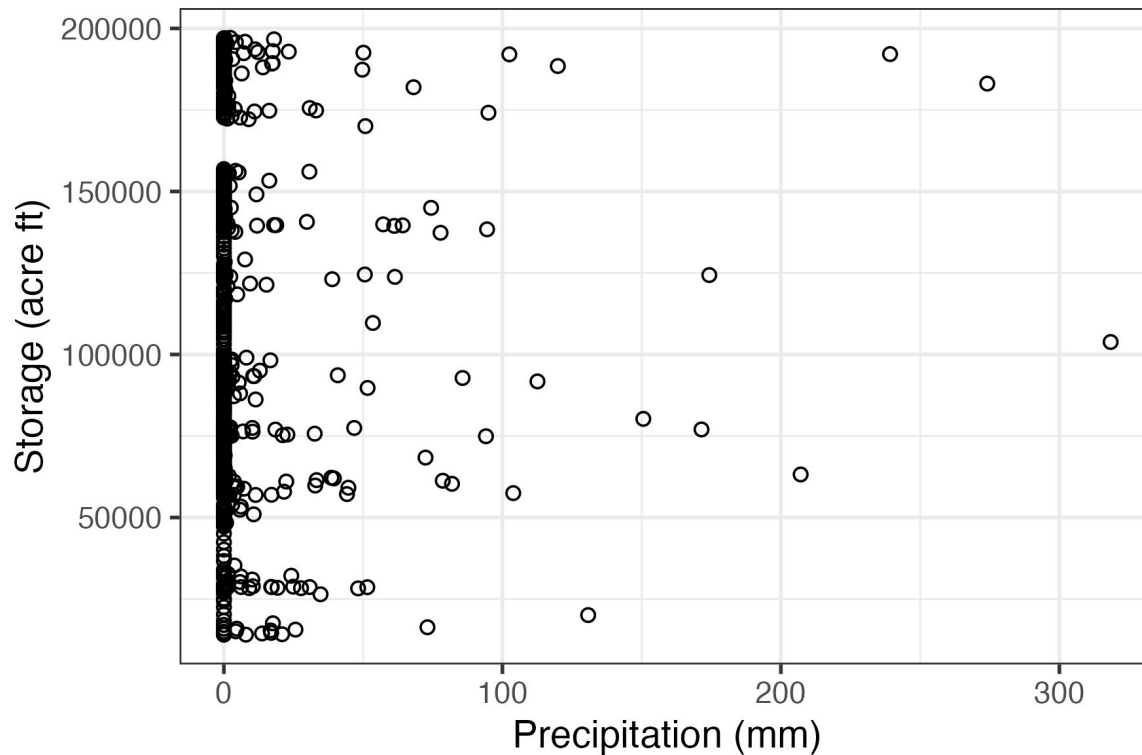
EDS 222: Statistics for Environmental Data Science



Lake Cachuma Reservoir



Lake Cachuma Reservoir



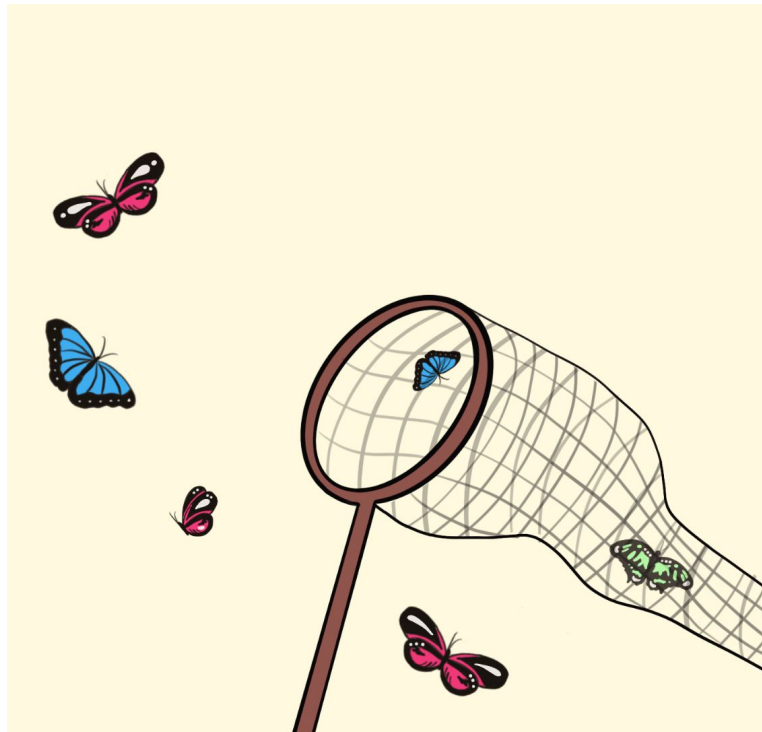
Today's agenda

- Correlation
- Temporal autocorrelation
- Spatial autocorrelation

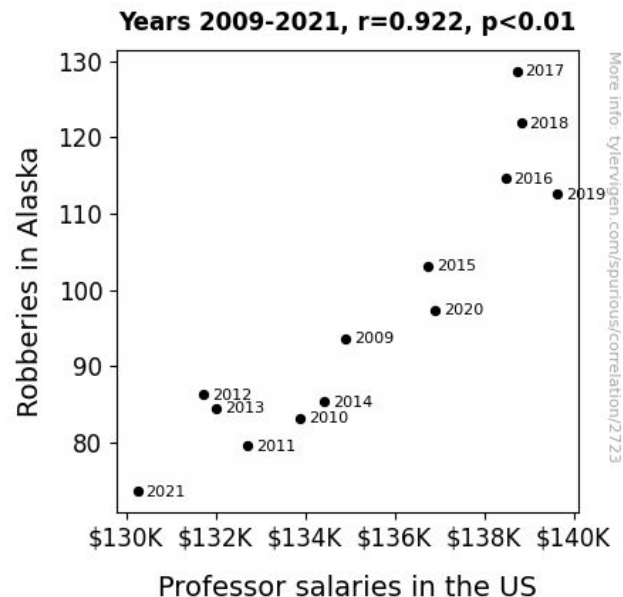


Today's agenda

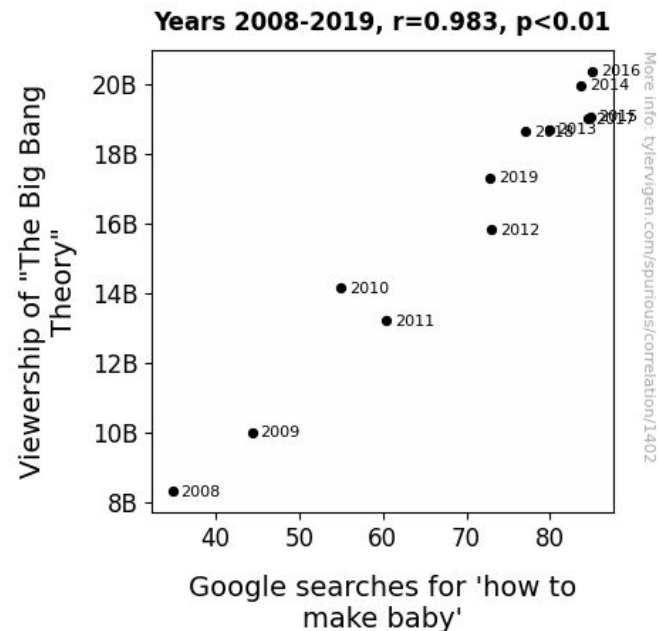
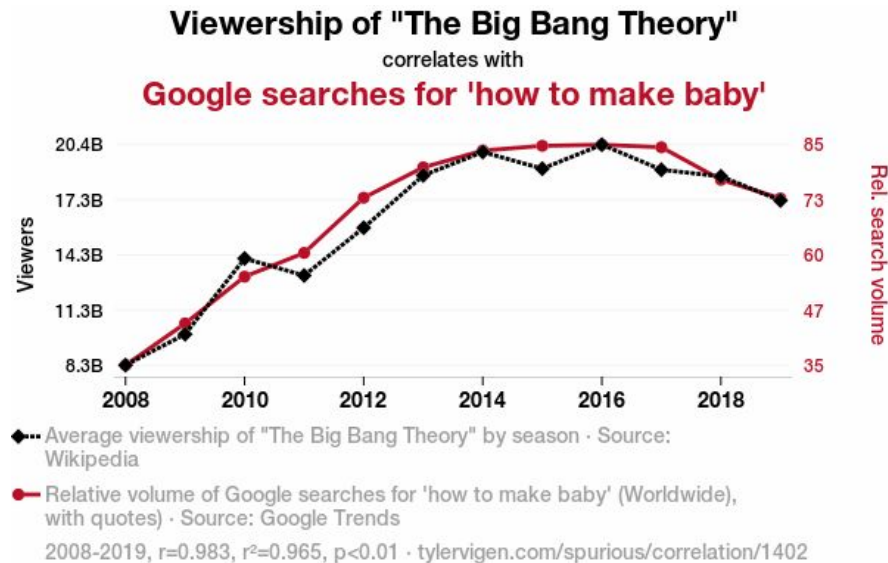
- **Correlation**
- Temporal autocorrelation
- Spatial autocorrelation



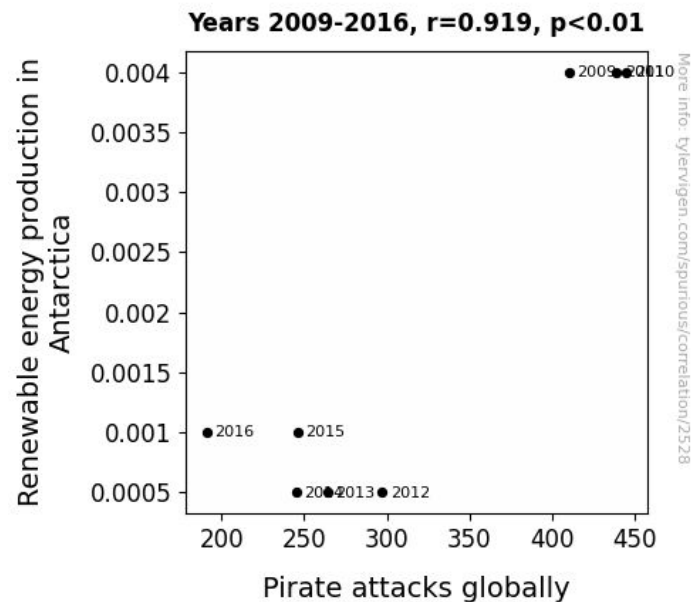
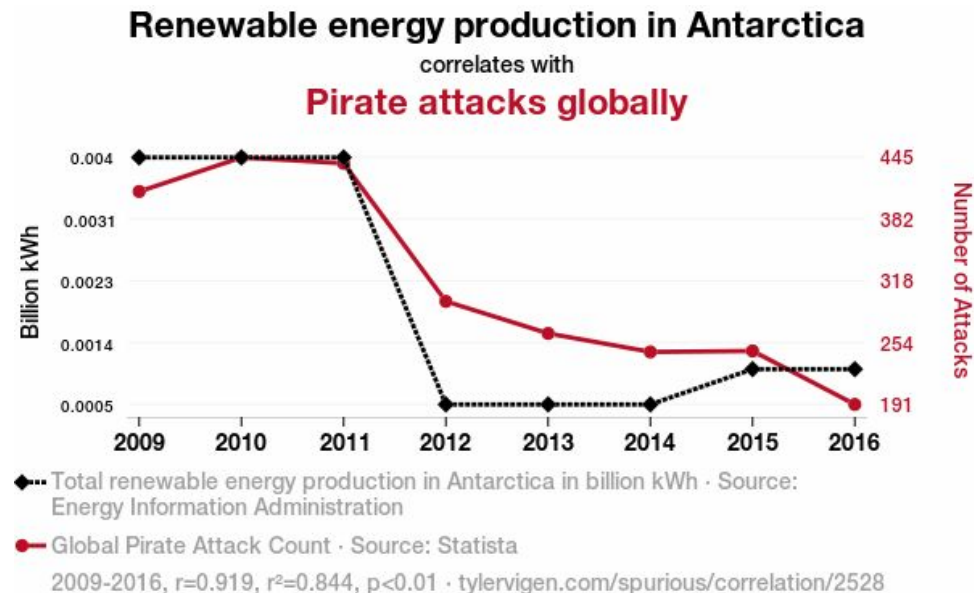
Correlations between variables



Correlations between variables



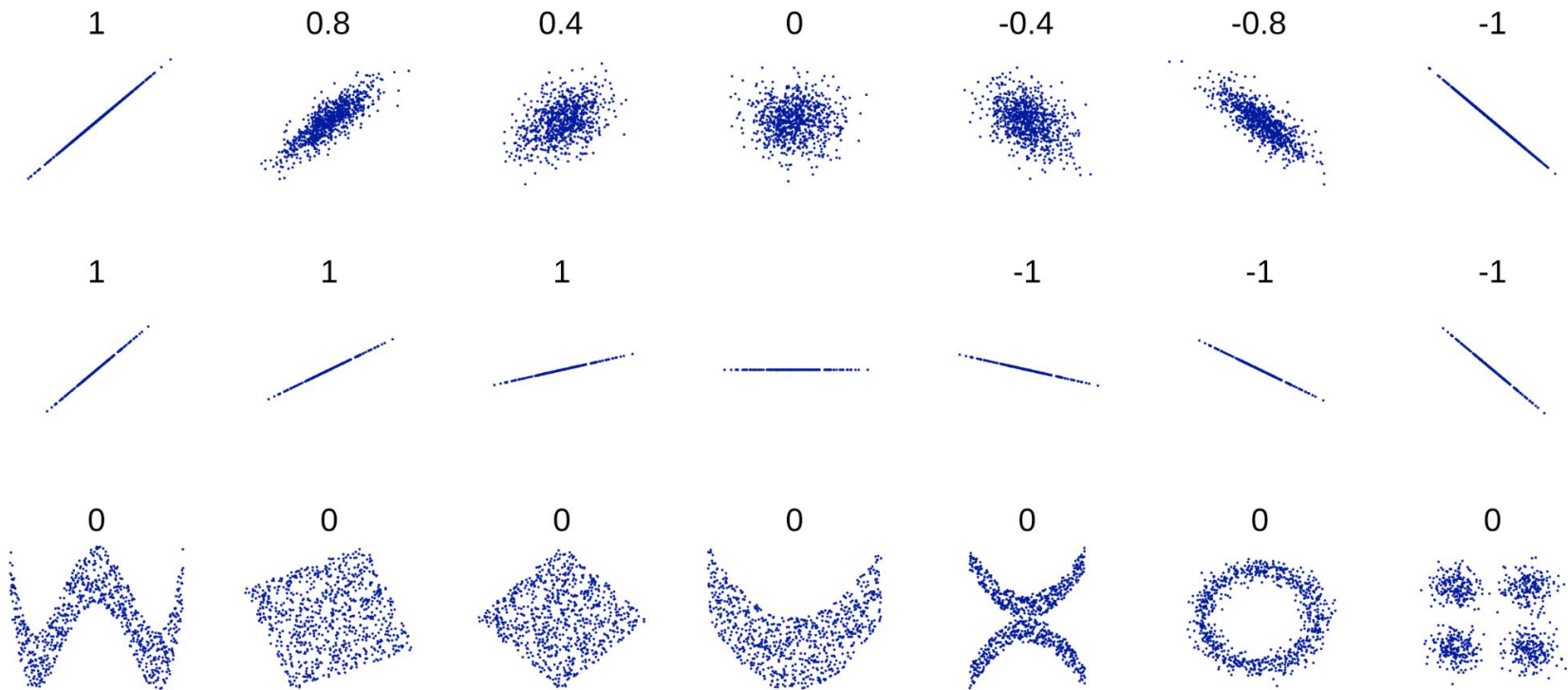
Correlations between variables



Coefficient of correlation

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

Coefficient of correlation

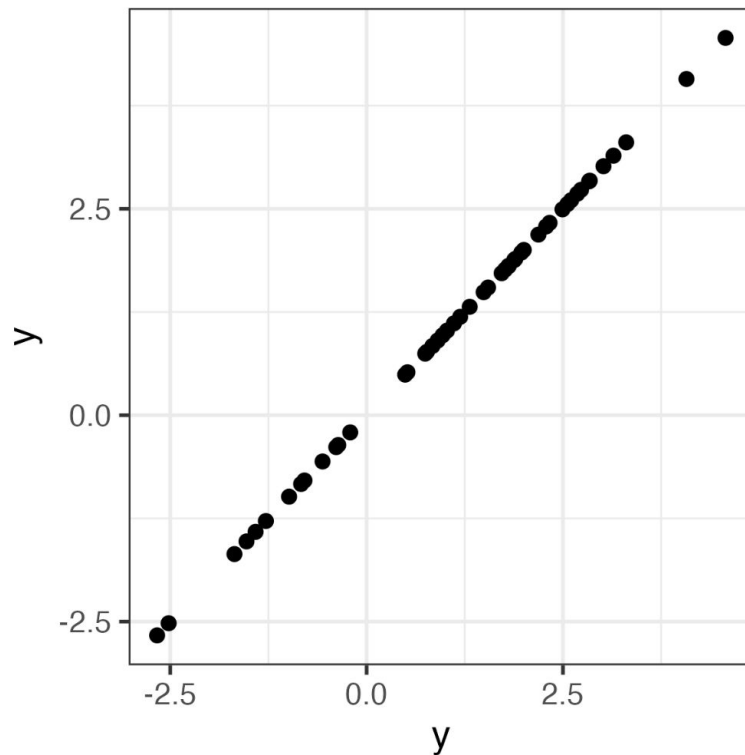
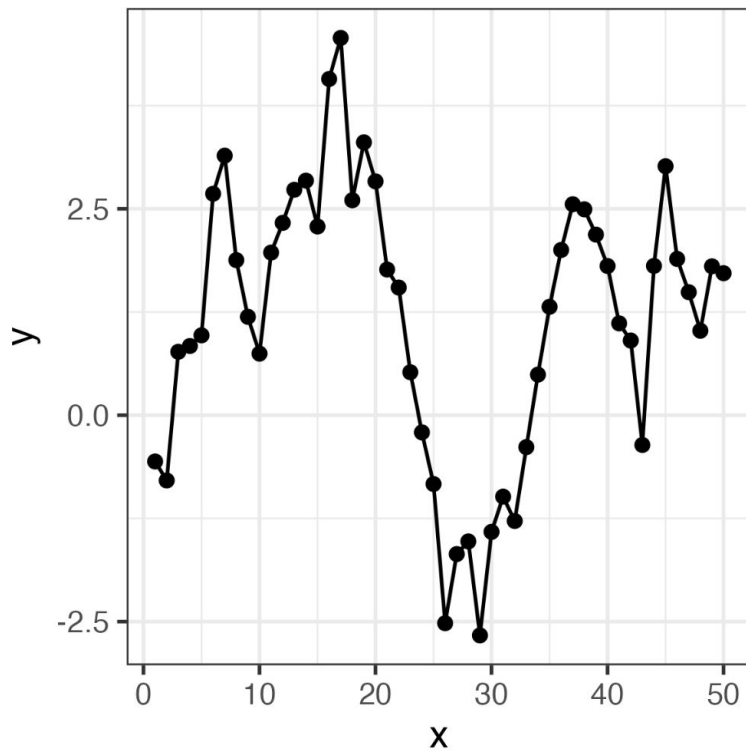


Today's agenda

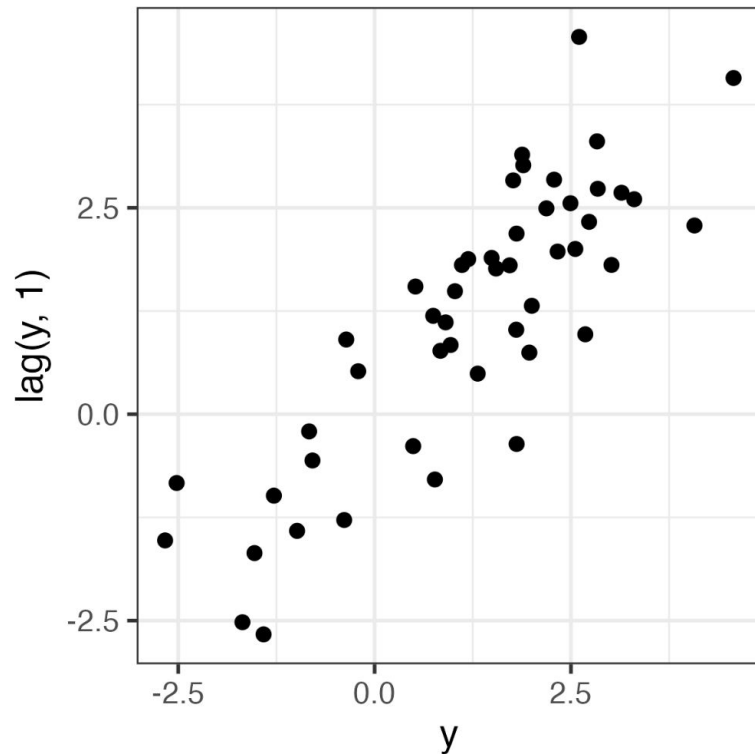
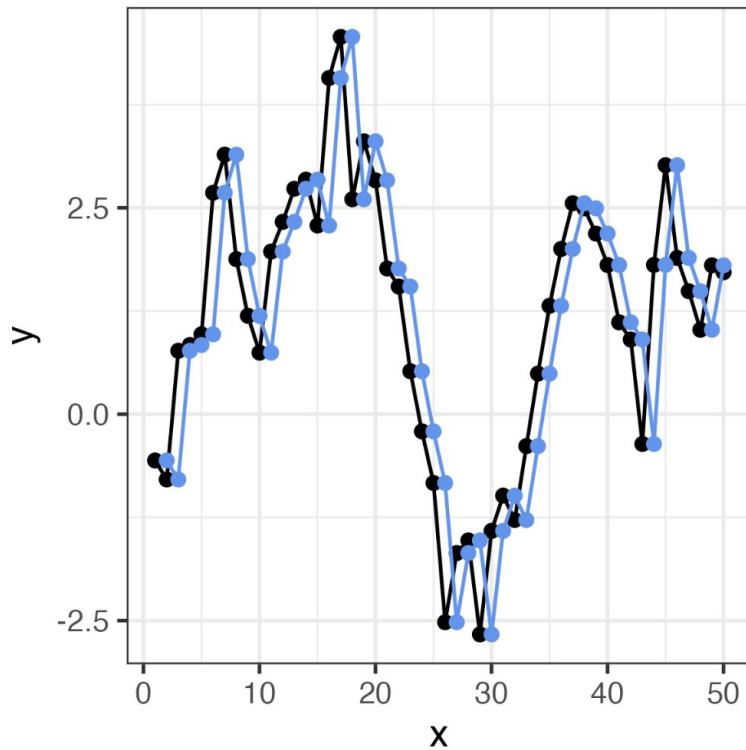
- Correlation
- **Temporal autocorrelation**
- Spatial autocorrelation



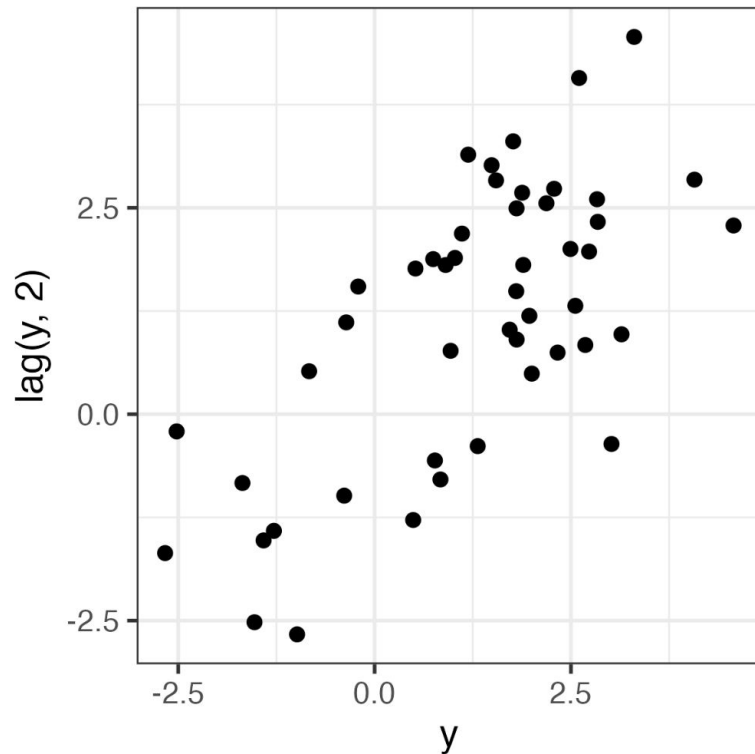
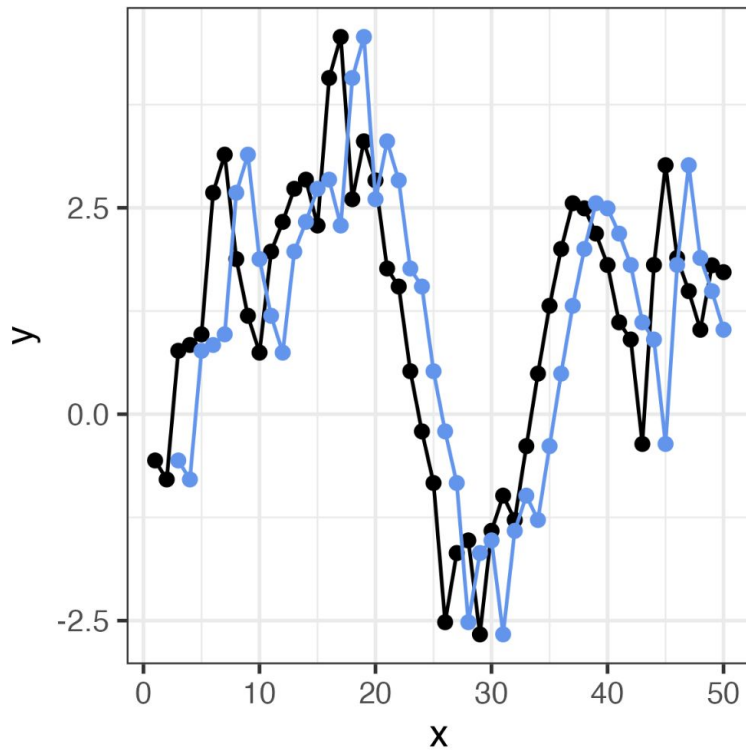
Temporal autocorrelation



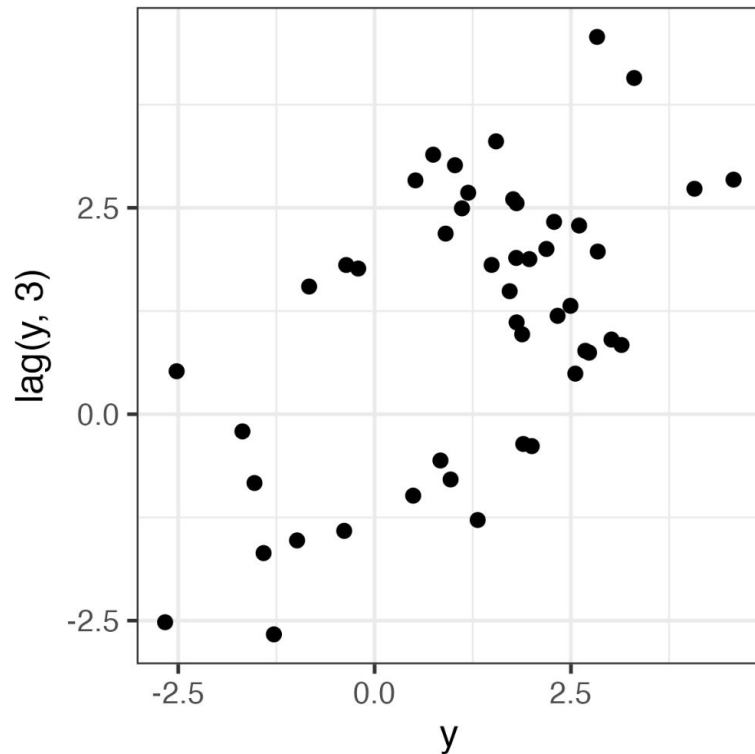
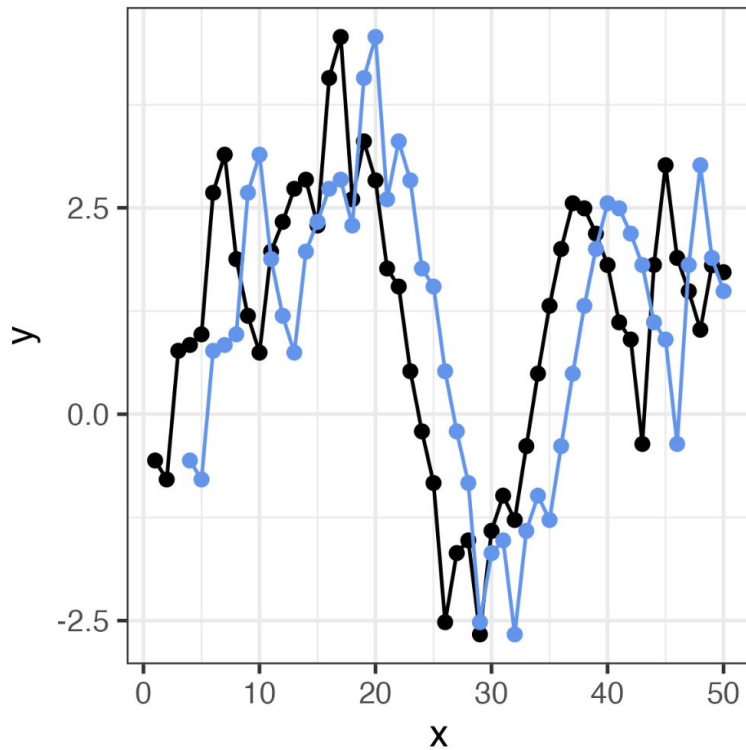
Temporal autocorrelation: lag=1



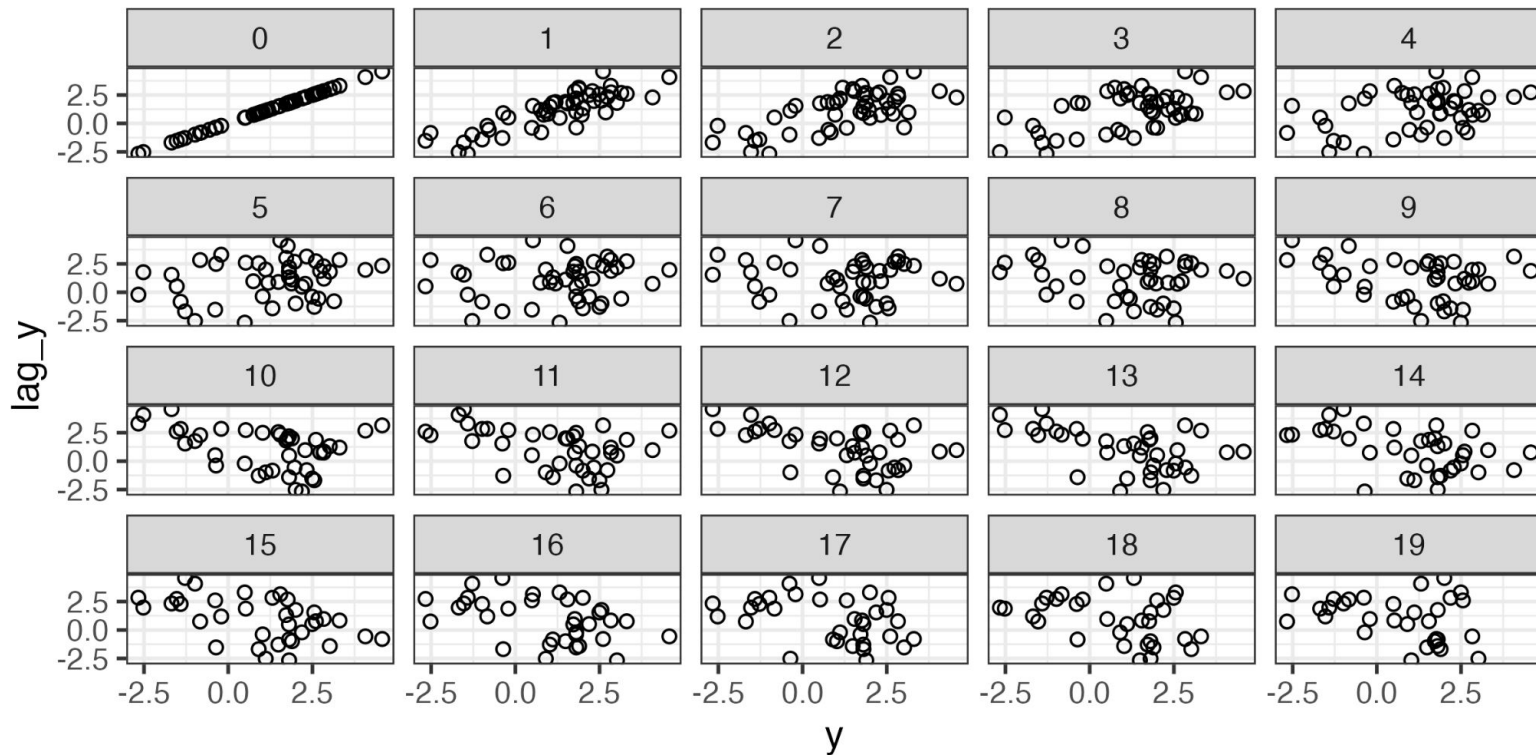
Temporal autocorrelation: lag=2



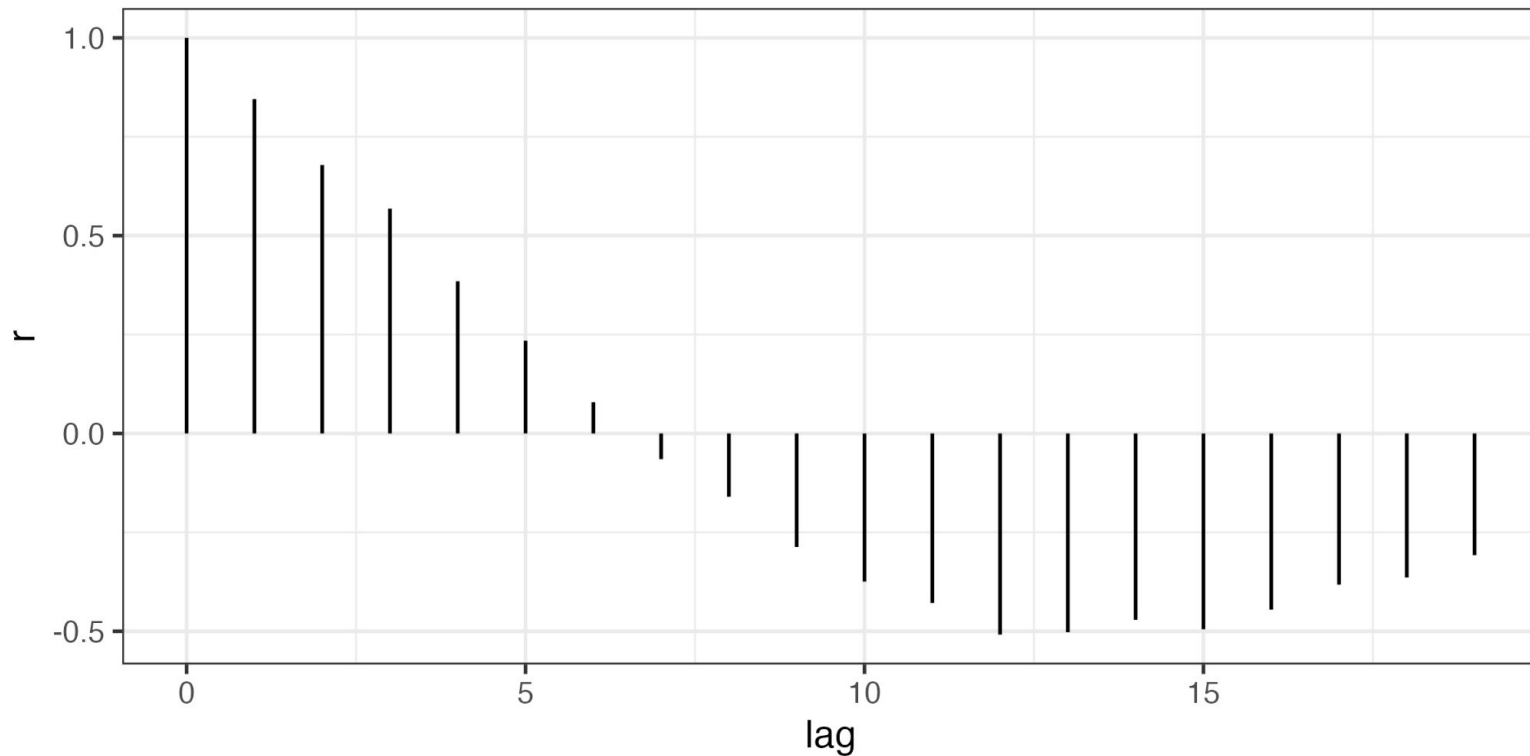
Temporal autocorrelation: lag=3



Temporal autocorrelation: lags 0-19



Temporal autocorrelation function



Temporal autocorrelation

Think-pair-share

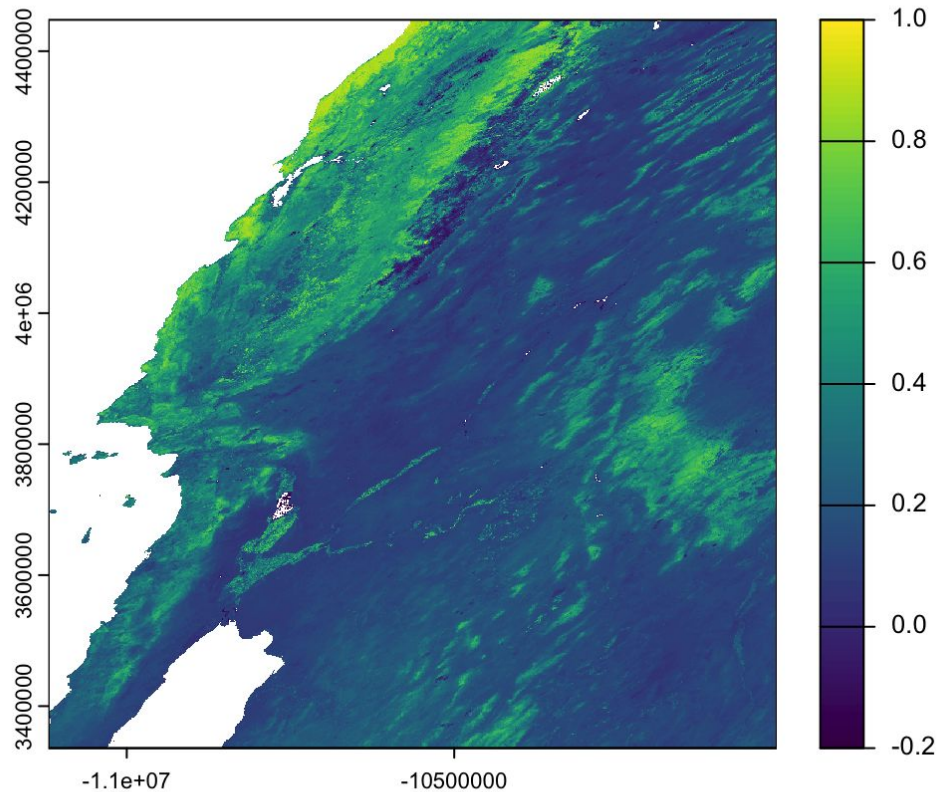
- What is an environmental variable that you think is temporally autocorrelated?
- How long do you think the autocorrelation persists? In other words, how many lags until the correlation approaches 0?

Today's agenda

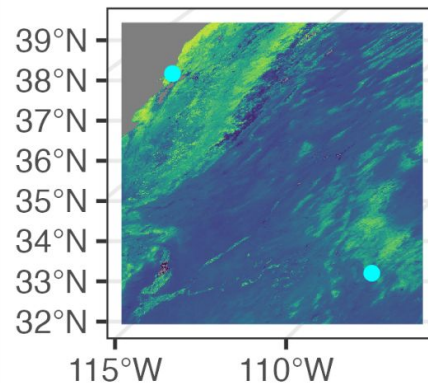
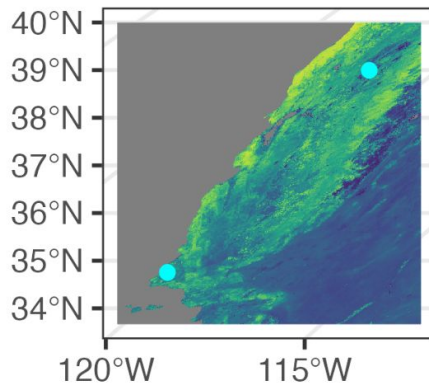
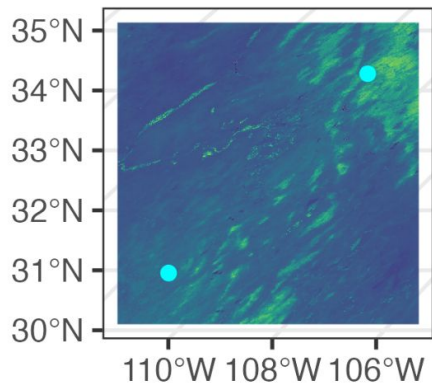
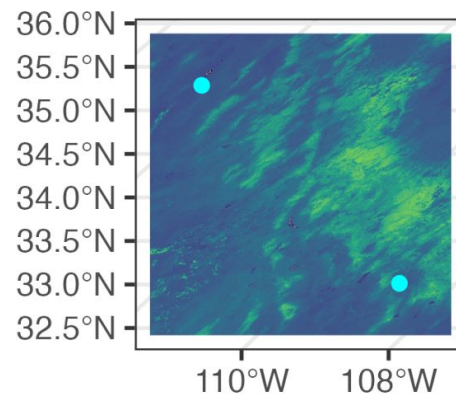
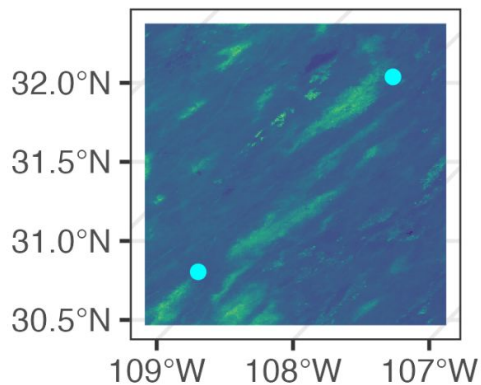
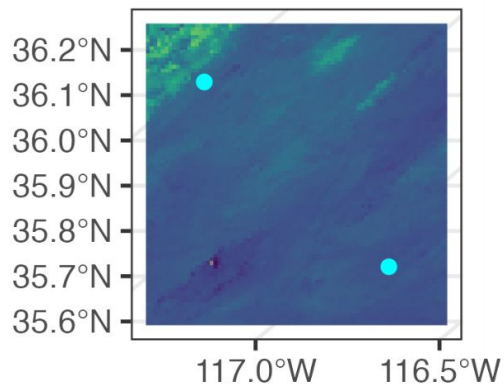
- Correlation
- Temporal autocorrelation
- **Spatial autocorrelation**



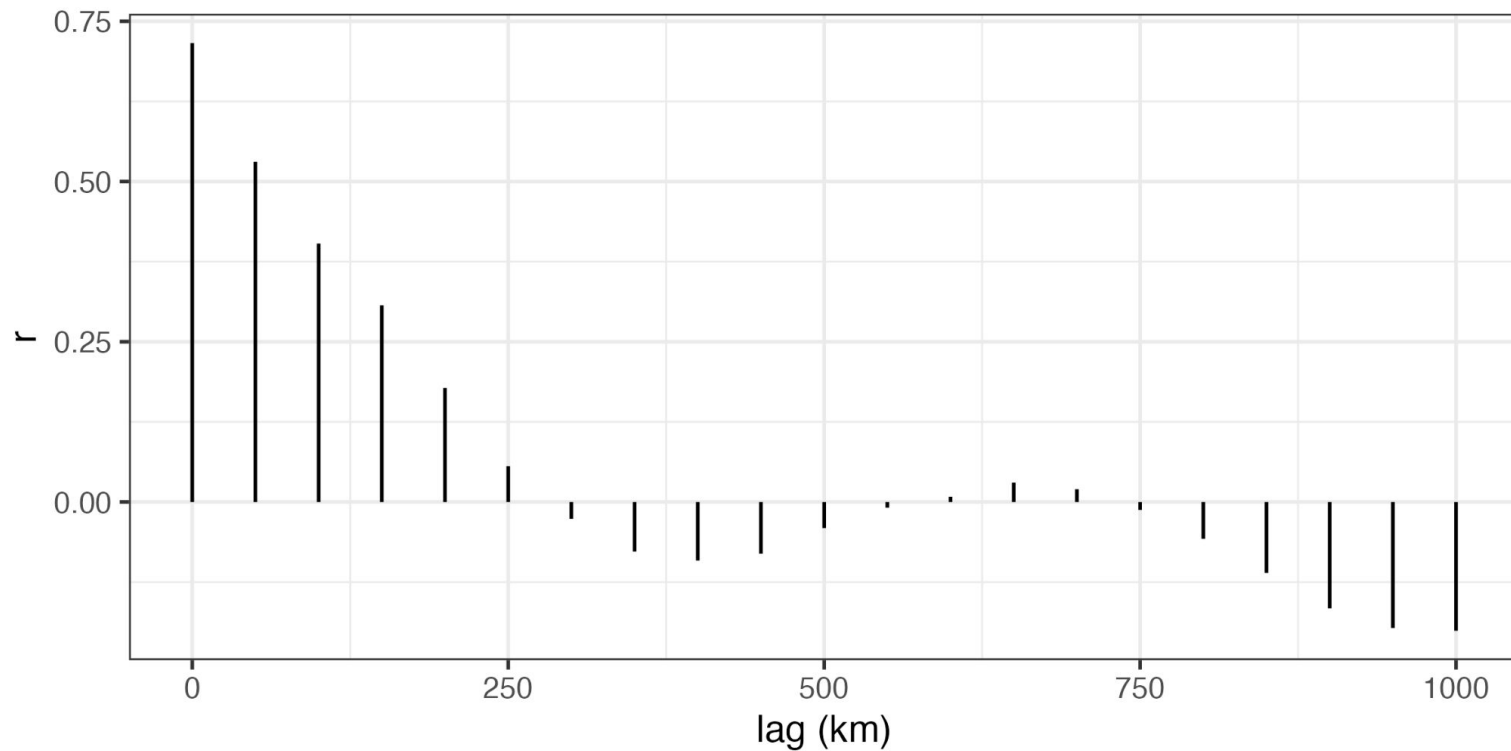
Spatial autocorrelation



Spatial autocorrelation



Spatial autocorrelation



Recap

- Autocorrelation is ubiquitous in environmental data
- Temporal autocorrelation: things closer in time are more similar
- Spatial autocorrelation: things closer in space are more similar
- Autocorrelation violates many of the assumptions we've grown accustomed to in regression!
- But it's not all bad - autocorrelation means observations contain information about their neighbors. Which we will take advantage of next week!